

Finite Fields

A field F that has finitely many elements is called a *finite field*.

There is a natural ring homomorphism $\mathbb{Z} \rightarrow F$ (which sends 1 to the multiplicative identity 1 in F). Since the image is a domain, the kernel is a prime ideal in $\mathbb{Z}$. Hence, there is a prime p in $\mathbb{Z}$ such that $\mathbb{Z}/\langle p \rangle$ is a subring of F .

Since $\mathbb{Z}/\langle p \rangle$ is a field and F is a finitely generated module over it, it is a finite dimensional vector space over this field. It follows that there is a positive integer n such that F has $q = p^n$ elements.

A finite field has $q = p^n$ elements, where n is a positive integer and p is a prime.

A finite field is a vector space over $\mathbb{Z}/\langle p \rangle$.

Multiplicative group

The non-zero elements of F form a commutative group F^* with $q - 1$ elements. It follows that all these elements are roots of the polynomial $X^{q-1} - 1$.

A polynomial of degree d over a field can have *at most* d distinct roots. In particular the polynomial $X^d - 1$ has at most d roots in F . If d is the least common multiple of the orders of *all* elements of F^* , then *every* element of F^* satisfies $X^d - 1$. So $d \geq (q - 1)$. Since d divides $q - 1$, we see that $d = q - 1$.

Given two elements a and b of an Abelian group of orders d and e , there is an element whose order is the least common multiple of d and e . Thus, the least common multiple of the orders of all elements of a finite group is the order of *some* element of the group.

Thus, there is an element α of F^* whose order is *exactly* $(q - 1)$.

The multiplicative group F^* of non-zero elements of a finite field F is a cyclic group.

Frobenius endomorphism

Given a commutative ring R with identity that contains $\mathbb{Z}/\langle p \rangle$ as a subring with the same identity. In this case we say that R is a $\mathbb{Z}/\langle p \rangle$ -algebra.

This means that $n \cdot a = 0$ for any a in R when n is an integer divisible by p .

By the Binomial theorem, we see that if a and b are elements of R , then

$$(a + b)^p = \sum_{r=0}^p \binom{p}{r} a^r \cdot b^{p-r} = a^p + b^p$$

since $\binom{p}{r}$ is divisible by p for $0 < r < p$.

Now, $(a \cdot b)^p = a^p \cdot b^p$ in any commutative ring R .

$\phi_p : R \rightarrow R$ defined by $\phi(a) = a^p$ is an endomorphism of an $\mathbb{Z}/\langle p \rangle$ -algebra R .

Given a positive integer n , we see that $\phi_q = (\phi_p)^n$ is also an endomorphism.

Note that if $\psi : R \rightarrow R$ is an endomorphism of a ring, then the set R^ψ of elements a of R that satisfy $\psi(a) = a$ is a subring.

Existence of a field with q elements

We now show that there *is* a field with q elements for any $q = p^n$ with p a prime and n a positive integer.

We have seen that non-zero elements of a finite field with q elements are roots of the polynomial $X^{q-1} - 1$. Hence, all the elements of the field (if it exists!) are roots of the polynomial $X^q - X$.

We calculate the discriminant in $\mathbb{Z}/\langle p \rangle$:

$$\begin{aligned} \text{Disc}(X^q - X) &= \text{Res}_{q,q-1}(X^q - X, -1) \\ &= (-1)^{q(q-1)} \text{Res}_{q-1,q}(-1, X^q - X) \\ &= 1^{q-1}(-1)^q \neq 0 \end{aligned}$$

since $\text{Syl}_{q,q-1}(-1, X^q - X)$ is an upper-triangular matrix with -1 as the first q diagonal entries and 1 as the last $q-1$ diagonal entries.

It follows that $X^q - X$ has *distinct* roots in some splitting field k containing $\mathbb{Z}/\langle p \rangle$.

Thus, the fixed subring $\mathbb{F}_q = k^{\phi_q}$ of k contains exactly q elements. Moreover, the non-zero elements are invertible since they satisfy $X^{q-1} - 1$. Hence, $\mathbb{F}_q$ is a field.

Minimal polynomial

Given $E \subset F$ fields such that F is a finite dimensional vector space over E ; let m be this dimension.

Given an element α of F we obtain a ring homomorphism $E[X] \rightarrow F$ which sends X to α . The image $E[\alpha]$ is a finite dimensional vector space of dimension at most m , we see that the kernel of this homomorphism contains a non-zero polynomial of degree at most m . The non-zero polynomial of least degree d that lies in the kernel with leading coefficient 1 is called the *minimal polynomial* of α over E .

Note that, by a simple application of long division, the minimal polynomial divides every other polynomial in the kernel of $E[X] \rightarrow E[\alpha]$.

Since $E[\alpha] \subset F$ is a domain, the kernel is generated by an irreducible polynomial of degree d . In other words, the minimal polynomial is an irreducible polynomial

of degree d . It follows that $E[\alpha]$ is a field. Clearly, it is the smallest subfield of F that contains E and α .

Existence of an irreducible polynomial of degree n

Given $q = p^n$ a power of a prime p , we have seen that there is a field $\mathbb{F}_q$ which has q elements.

We have seen that there is an element α in $\mathbb{F}_q$ which has multiplicative order $p^n - 1$. Suppose $\mathbf{a}$ is the minimal polynomial of α over $\mathbb{Z}/\langle p \rangle$.

We have seen that $(\mathbb{Z}/\langle p \rangle)[\alpha]$ is a vector space of dimension $m \leq n$ over $\mathbb{Z}/\langle p \rangle$; hence, it has p^m elements.

Since $(\mathbb{Z}/\langle p \rangle)[\alpha]$ is a field, every non-zero element has order dividing $p^m - 1$. In particular, α has order dividing $p^m - 1$. Since α has order $p^n - 1$ and $m \leq n$, it follows that $m = n$.

There is an irreducible polynomial $\mathbf{a}$ of degree n over $\mathbb{Z}/\langle p \rangle$.

The field $\mathbb{F}_q$ can be identified with $((\mathbb{Z}/\langle p \rangle)[X])/\langle \mathbf{a} \rangle$.

Note that the irreducible polynomial $\mathbf{a}$ of degree n need not be *unique*!

Uniqueness upto isomorphism

Since α satisfies $X^{q-1} - 1$, we see that $\mathbf{a}$ is an irreducible polynomial in $(\mathbb{Z}/\langle p \rangle)[X]$ which divides $X^q - X$.

If F is *any* field with $q = p^n$ elements, we have seen that $X^q - X$ splits completely in F . It follows that if $\mathbf{b}$ is *any* polynomial which divides $X^q - X$, then $\mathbf{b}$ splits completely in F .

In particular, we see that there is an element β in F which is a root of the irreducible polynomial $\mathbf{a}$ of degree n . As above, we get a homomorphism $(\mathbb{Z}/\langle p \rangle)[X] \rightarrow F$ which sends X to β and the kernel of this is generated by $\mathbf{a}$.

This gives get a natural injective homomorphism $((\mathbb{Z}/\langle p \rangle)[X])/\langle \mathbf{a} \rangle \rightarrow F$. Since both sides have the same number of elements, this is an isomorphism.

Up to isomorphism, there is a *unique* field $\mathbb{F}_q$ with q elements for every prime power q .

Reciprocity

The field $\mathbb{F}_q$ contains a primitive $(q - 1)$ -th root of unity α for $q = p^n$ a power of a prime p .

Given an integer k which divides $p^n - 1$, we see that $\mathbb{F}_{p^n}$ contains a primitive d -th root of unity (the element $\alpha^{(q-1)/k}$ is such a root of unity).

Thus, the *smallest* n for which $\mathbb{F}_{p^n}$ contains a primitive k -th root of unity is also the smallest n for which k divides $p^n - 1$.

If k divides $p^n - 1$, then k and p are co-prime. Thus, p is an element of the multiplicative group $(\mathbb{Z}/\langle k \rangle)^*$ of invertible elements in $\mathbb{Z}/\langle k \rangle$.

We see that the smallest n as above is the *order* of p as an element of this group.

The smallest n such that $\mathbb{F}_{p^n}$ contains a primitive k -th root of unity is the order of p as an element of $(\mathbb{Z}/\langle k \rangle)^*$.

This is called (*cyclotomic*) *reciprocity* since the problem of solving $X^k - 1$ modulo p is reduced to the behaviour of p in $\mathbb{Z}/\langle k \rangle$.

Its immediate use is that we can work with p quite large as long as k is not too big since we only need to look at $p \bmod k$.